LEVEL 1 — GENERAL NORMALIZED EXPRESSION

$$MP_{norm} = [(MI \times NP) \times SW \times MS^{\gamma} \times CP^{\alpha}]^{1/n_{eff}}$$

$n_{eff} = 3 + \gamma + \alpha$ $\quad MP_{norm} \in (0, 1]$ $\quad$ all components $\in (0, 1]$

Weighted geometric mean with weights $\omega_{MI} = \omega_{NP} = \omega_{SW} = 1/n_{eff}$, $\omega_{MS} = \gamma/n_{eff}$, $\omega_{CP} = \alpha/n_{eff}$

γ: weight of the security environment $\quad$ α: weight of cyber dependence, not a measure of capability

LEVEL 2 — COMPONENT DISAGGREGATION

MI

Maritime Interests

What the state seeks to obtain, protect, or project through the sea.

$$MI = \Sigma_i \, w_i \cdot MI_i$$

$\Sigma \, w_i = 1$; $MI \in (0, 1]$

Weighted linear sub-index: interests are substitutable and w_i act as rates of substitution.

Colombia: n = 18 maritime interests.

NP

Naval Power

Capacity to act on the maritime domain to defend interests, deter threats, and project influence.

$$NP = F \times P \times MDA^{\mu}$$

$NP \in (0, 1]$; μ: MDA calibration

F: force, including tactical-operational maritime cyberdefense.

P: position. MDA: coverage and quality of the maritime picture.

SW

Strategic Will

Capacity to turn maritime interests into sustained decisions, capabilities, and actions.

$$SW = (PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3})^{1/\Sigma w}$$

$SW \in (0, 1]$; multiplier of the whole system

PW: political, IW: institutional, $SocW$: social will.

Maritime consciousness, a cultural construct distinct from MDA, enters through $SocW$.

MS

Maritime Security

Condition of the strategic environment: it measures the incidence of threats, not capabilities.

$$MS = (IC^{a_1} \cdot TE^{a_2} \cdot BC^{a_3})^{1/\Sigma a}$$

$MS \in (0, 1]$; global weight γ

IC: interstate conflicts and gray zone. TE: terrorism and extremism. BC: blue crime.

Cyber vectors of each threat count here.

CP

Cyber Power

Capacity to protect, govern, and restore the cyber assets of the maritime domain, amplified by diplomatic projection.

$$CP = \min\{1, (\beta_1 \, CS_m + \beta_2 \, CD_m + \beta_3 \, RC_m)(1 + \delta \cdot CDip)\}$$

$CP \in (0, 1]$; $\Sigma \, \beta_i = 1$; global weight α

CS_m: maritime cybersecurity. CD_m: strategic maritime cyberdefense.

RC_m: maritime recovery capacity, $\min\{1, t_{ref}/t\}$. $CDip$: cyber diplomacy, amplifier only.

OPERATIONAL BOUNDARIES BETWEEN COMPONENTS

- **Cyberdefense:** execution of operations (active defense, military cyber intelligence, offensive capabilities) in F; governance, cooperation, doctrine, and deterrent posture in CD_m.
- **Protection:** controls and information sharing over maritime IT/OT assets in CS_m; restoration after an incident in RC_m.
- **Awareness:** MDA measures the maritime picture; the protection of the systems that produce it is measured in CS_m.
- **Cyber diplomacy:** enters only through $CDip$; excluded from PW and from defense cooperation in CD_m.
- **Threats:** their incidence, including cyber vectors, only in MS; capabilities against them only in NP and CP.